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ABSTRACT.....4

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This study develops a reproducible machine-learning workflow for red wine quality prediction from physicochemical measurements. The dataset contains 1,599 samples and eleven input variables. Following the assignment specification, the first 1,300 samples are used for training and the remaining 299 samples are held out for final testing. All preprocessing, feature selection, principal component analysis, model comparison, and hyperparameter search are conducted on the training set only. Logistic regression, support vector machine, random forest, ExtraTrees, and histogram gradient boosting are compared with five-fold stratified cross-validation. Because the quality classes are imbalanced, macro-F1 is used as the primary selection metric alongside accuracy, balanced accuracy, weighted-F1, and MAE. The selected SelectKBest plus ExtraTrees pipeline obtains a best cross-validation macro-F1 of 0.4529. On the held-out test set, it achieves an accuracy of 0.5518, a macro-F1 of 0.2776, and an MAE of 0.5318. The results show reasonable discrimination for the dominant quality levels while also revealing the difficulty of predicting very rare quality scores.

Key words: red wine quality prediction; machine learning; feature selection; ExtraTrees; cross-validation

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